

Practical - 4

Aim: Create Node.js application with MySQL database

1: Write a node.js code to establish connection with MySQL and create database “Node_test”.

Code:

```
const mysql = require('mysql2');

// Connect to MySQL
const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
});

con.connect((err) => {
  if (err) throw err;

  console.log('Connected to MySQL!');

  // Create database
  con.query('CREATE DATABASE IF NOT EXISTS Node_test', (err) => {
    if (err) throw err;

    console.log("Database 'Node_test' created successfully.");
    con.end();
  });
});
```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_1.js
Connected to MySQL!
Database 'Node_test' created successfully.
```

2: Write a node.js code to create a table ‘Product’ with Id, Name, Brand, Quantity and Price as fields.

Code:

```
const mysql = require('mysql2');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
```

```
});

con.connect((err) => {
  if (err) throw err;

  console.log('Connected to Node_test database!');
  const sql = `
    CREATE TABLE IF NOT EXISTS Product (
      Id INT AUTO_INCREMENT PRIMARY KEY,
      Name VARCHAR(100),
      Brand VARCHAR(100),
      Quantity INT,
      Price DECIMAL(10,2)
    )
  `;
  con.query(sql, (err) => {
    if (err) throw err;
    console.log('Product table created successfully.');
```

```
con.end();
  });
});
```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_2.js
Connected to Node_test database!
Product table created successfully.
```

3: Write a node.js program that takes product information like Name, Brand, Quantity and Price from the user using HTML form and stores it into the Product table as mentioned below:

Product				
Id	Name	Brand	Quantity	Price
1	Soap	Dettol	3	20
2	Shampoo	Loreal	5	100
3	Toothpaste	Colgate	2	50
4	Sanitizer	LifeBoy	2	60

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');
```

```
const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);

  // Display form
  if (url.pathname === '/') {
    res.writeHead(200, { 'Content-Type': 'text/html' });

    res.end(`
      <h1>Add Product</h1>
      <form action="/add" method="GET">
        Name:
        <input type="text" name="name" required>
        <br><br>
        Brand:
        <input type="text" name="brand" required>
        <br><br>
        Quantity:
        <input type="number" name="quantity" required>
        <br><br>
        Price:
        <input type="number" name="price" required>
        <br><br>
        <button type="submit">Add Product</button>
      </form>
    `);
  }

  // Insert product
  else if (url.pathname === '/add') {
    const name = url.searchParams.get('name');
    const brand = url.searchParams.get('brand');
    const quantity = url.searchParams.get('quantity');
    const price = url.searchParams.get('price');

    const sql = `
      INSERT INTO Product (Name, Brand, Quantity, Price)
      VALUES (?, ?, ?, ?)
    `;

    con.query(sql, [name, brand, quantity, price], (err) => {
      if (err) {
        res.writeHead(500);
        res.end('Database Error');
      }
    });
  }
});
```

```

        return;
    }
    res.writeHead(200, {
        'Content-Type': 'text/html',
    });
    res.end(`
        <h2>Product added successfully!</h2>
        <a href="/">Add another product</a>
    `);
});
}
});

server.listen(3000, () => {
    console.log('Server running at http://localhost:3000');
});

```

Output:

```

C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_3.js
Server running at http://localhost:3000

```

The image shows two screenshots of a web browser. The first screenshot shows a form titled "Add Product" with fields for Name (Soap), Brand (Dettol), Quantity (3), and Price (20), and an "Add Product" button. The second screenshot shows the browser at the URL "localhost:3000/add?name=Soap&brand=Dettol&quantity=3&price=20", displaying the message "Product added successfully!" and a link "Add another product".

4: Write a node.js program to retrieve all the details of the product and display in browser.

Code:

```

const http = require('http');
const mysql = require('mysql2');

const con = mysql.createConnection({
    host: 'localhost',
    user: 'vatsal',
    password: 'vatsal',

```

```

    database: 'Node_test',
  });
  const server = http.createServer((req, res) => {
    if (req.url === '/products') {
      const sql = 'SELECT * FROM Product';
      con.query(sql, (err, result) => {
        if (err) {
          res.writeHead(500);
          res.end('Database Error');
          return;
        }
        let html = `
          <html>
          <head>
            <title>Products</title>
            <style>
              table {
                border-collapse: collapse;
                width: 70%;
              }
              th, td {
                border: 1px solid black;
                padding: 10px;
                text-align: center;
              }
              th {
                background-color: lightgray;
              }
            </style>
          </head>
          <body>
            <h1>Product Details</h1>
            <table>
              <tr>
                <th>Id</th>
                <th>Name</th>
                <th>Brand</th>
                <th>Quantity</th>
                <th>Price</th>
              </tr>
            </table>
          `;

        result.forEach((product) => {
          html += `
            <tr>
              <td>${product.Id}</td>
              <td>${product.Name}</td>
              <td>${product.Brand}</td>
              <td>${product.Quantity}</td>
              <td>${product.Price}</td>
            </tr>
          `;
        });
      });
    }
  });

```

```

        </tr>
    `;
    });
    html += `
        </table>
    </body>
    </html>
    `;
    res.writeHead(200, {
        'Content-Type': 'text/html',
    });
    res.end(html);
    });
} else {
    res.writeHead(404);
    res.end('404 - Page Not Found');
}
});

server.listen(3000, () => {
    console.log('Server running at http://localhost:3000/products');
});

```

Output:

```

C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_4.js
Server running at http://localhost:3000/products

```

← → ↻ http://localhost:3000/products

Product Details

Id	Name	Brand	Quantity	Price
1	Soap	Dettol	3	20.00
2	Shampoo	Loreal	5	100.00
3	Toothpaste	Colgate	2	50.00
4	Sanitizer	LifeBoy	2	60.00

5: Write a node.js program to search any product by Name or Brand.

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);
  if (url.pathname === '/') {
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(`
      <h1>Search Product</h1>
      <form action="/search" method="GET">
        Enter Name or Brand:
        <input type="text" name="value" required>
        <button type="submit">Search</button>
      </form>
    `);
  } else if (url.pathname === '/search') {
    const value = url.searchParams.get('value');
    const sql = `
      SELECT * FROM Product
      WHERE Name LIKE ? OR Brand LIKE ?
    `;
    const searchValue = `%${value}%`;
    con.query(sql, [searchValue, searchValue], (err, result) => {
      if (err) {
        res.writeHead(500);
        res.end('Database Error');
        return;
      }
      let html = `
        <h1>Search Results</h1>
        <table border="1" cellpadding="10">
          <tr>
            <th>Id</th>
            <th>Name</th>
            <th>Brand</th>
            <th>Quantity</th>
            <th>Price</th>
          </tr>
        `;
    });
  }
});
```

```

    `;
    result.forEach((product) => {
      html += `
        <tr>
          <td>${product.Id}</td>
          <td>${product.Name}</td>
          <td>${product.Brand}</td>
          <td>${product.Quantity}</td>
          <td>${product.Price}</td>
        </tr>
      `;
    });
    html += '</table>';
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(html);
  });
} else {
  res.writeHead(404);
  res.end('404 - Page Not Found');
}
});
server.listen(3000, () => {
  console.log('Server running at http://localhost:3000');
});

```

Output:

```

C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_5.js
Server running at http://localhost:3000

```

← → ↻ http://localhost:3000/

Search Product

Enter Name or Brand:

← → ↻ http://localhost:3000/search?value=Loreal

Search Results

Id	Name	Brand	Quantity	Price
2	Shampoo	Loreal	5	100.00

6: Write a node.js program to retrieve all the products whose quantity is less than 3.

Code A:

```
const http = require('http');
const mysql = require('mysql2');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  if (req.url === '/products') {
    const sql = 'SELECT * FROM Product WHERE Quantity < 3';
    con.query(sql, (err, result) => {
      if (err) {
        res.writeHead(500);
        res.end('Database Error');
        return;
      }
      let html = `
        <h1>Products With Quantity Less Than 3</h1>
        <table border="1" cellpadding="10">
          <tr>
            <th>Id</th>
            <th>Name</th>
            <th>Brand</th>
            <th>Quantity</th>
            <th>Price</th>
          </tr>
      `;
      result.forEach((product) => {
        html += `
          <tr>
            <td>${product.Id}</td>
            <td>${product.Name}</td>
            <td>${product.Brand}</td>
            <td>${product.Quantity}</td>
            <td>${product.Price}</td>
          </tr>
        `;
      });
      html += '</table>';
      res.writeHead(200, {
        'Content-Type': 'text/html',
      });
      res.end(html);
    });
  } else {
```

```
    res.writeHead(404);
    res.end('404 - Page Not Found');
  }
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000/products');
});
```

Output A:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_6.js
Server running at http://localhost:3000/products
```

← → ↻ http://localhost:3000/products

Products With Quantity Less Than 3

Id	Name	Brand	Quantity	Price
3	Toothpaste	Colgate	2	50.00
4	Sanitizer	LifeBoy	2	60.00

7: Write a node.js program to update price of a specific item in the product table.

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);
  // Display form
  if (url.pathname === '/') {
    res.writeHead(200, {
      'Content-Type': 'text/html',
```

```

});
res.end(`
  <h1>Update Product Price</h1>
  <form action="/update" method="GET">
    Product ID:
    <input type="number" name="id" required>
    <br><br>
    New Price:
    <input type="number" name="price" required>
    <br><br>
    <button type="submit">Update Price</button>
  </form>
`);
}
// Update price
else if (url.pathname === '/update') {
  const id = url.searchParams.get('id');
  const price = url.searchParams.get('price');

  const sql = `
    UPDATE Product
    SET Price = ?
    WHERE Id = ?
  `;

  con.query(sql, [price, id], (err, result) => {
    if (err) {
      res.writeHead(500);
      res.end('Database Error');
      return;
    }
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(`
      <h2>Product price updated successfully.</h2>
      <p>Product ID: ${id}</p>
      <p>New Price: ${price}</p>

      <a href="/">Go Back</a>
    `);
  });
} else {
  res.writeHead(404);
  res.end('404 - Page Not Found');
}
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000');
});

```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_7.js
Server running at http://localhost:3000
```

← → ↻ http://localhost:3000/

Update Product Price

Product ID:

New Price:

← → ↻ http://localhost:3000/update?id=3&price=70

Product price updated successfully.

Product ID: 3

New Price: 70

[Go Back](#)

8: Write a node.js program to delete specific product from the product table.

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);

  // Display form
  if (url.pathname === '/') {
    res.writeHead(200, {
```

```

    'Content-Type': 'text/html',
  });

  res.end(`
    <h1>Delete Product</h1>

    <form action="/delete" method="GET">

      Product ID:
      <input type="number" name="id" required>

      <button type="submit">Delete</button>

    </form>
  `);
}

// Delete product
else if (url.pathname === '/delete') {
  const id = url.searchParams.get('id');

  const sql = 'DELETE FROM Product WHERE Id = ?';
  con.query(sql, [id], (err, result) => {
    if (err) {
      res.writeHead(500);
      res.end('Database Error');
      return;
    }
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(`
      <h2>Product deleted successfully.</h2>

      <p>Deleted Product ID: ${id}</p>

      <a href="/">Go Back</a>
    `);
  });
} else {
  res.writeHead(404);
  res.end('404 - Page Not Found');
}
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000');
});

```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_8.js
Server running at http://localhost:3000
```

← → ↻ http://localhost:3000/

Delete Product

Product ID:

← → ↻ http://localhost:3000/delete?id=4

Product deleted successfully.

Deleted Product ID: 4

[Go Back](#)